

Advanced (Habanero)**Q1.** What is the point-slope form of a linear equation?

(A) $y - y_1 = m(x - x_1)$

(B) $y = mx + b$

(C) $Ax + By = C$

(D) $x = my + b$

(E) $y = \frac{1}{m}x + b$

(A) 2

(B) -2

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

(E) 1

Q2. Which of the following is the slope-intercept form of a linear equation?

(A) $y - y_1 = m(x - x_1)$

(B) $y = mx + b$

(C) $Ax + By = C$

(D) $x = my + b$

(E) $y = \frac{1}{m}x + b$

Q6. The slope of the line passing through points (2, 3) and (4, 5) is

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

(E) $\frac{5}{2}$

Q3. What is the standard form of the equation $2y = 3x - 5$?

(A) $2y = 3x - 5$

(B) $3x - 2y = 5$

(C) $3x - 2y - 5 = 0$

(D) $2y - 3x = 5$

(E) $2y + 3x = 5$

Q7. The equation $x + 2y = 6$ can be rewritten in slope-intercept form as

(A) $y = \frac{1}{2}x + 3$

(B) $y = -\frac{1}{2}x + 3$

(C) $y = \frac{1}{2}x - 3$

(D) $y = -\frac{1}{2}x - 3$

(E) $y = \frac{3}{2}x - 1$

Q4. The equation $y = 2x - 3$ has a y -intercept of

(A) $(-3, 0)$

(B) $(0, -3)$

(C) $(3, 0)$

(D) $(0, 2)$

(E) $(2, 0)$

Q8. The graphs of $y = 2x - 3$ and $y = -2x + 1$ intersect at the point

(A) $(\frac{2}{2}, \frac{-2}{2})$

(B) $(\frac{2}{2}, \frac{2}{2})$

(C) $(\frac{1}{2}, 0)$

(D) $(\frac{1}{2}, \frac{1}{2})$

(E) $(\frac{4}{4}, \frac{4}{4})$

Q5. The equation $y = -2x + 1$ has a slope of**Open Ended Questions (FRQ)****Q9.** Graph the equation $y = -x + 2$ and find the x -intercept. Show your work.**Q10.** Write the equation of the line passing through (1, 2) with slope $m = -1$. Then convert the equation to standard form. Show your work.**Chef's Tips**

- Tip 1: Read each question carefully before selecting an answer.
- Tip 2: Use the formulas for point-slope, slope-intercept, and standard forms to help you answer questions.
- Tip 3: Graphing the equations can help you visualize the problems.